

Addendum: brute-force OR sweep at $K = 1$

Does the canonical $\pi/4$ area $K = 1$ admit any $\overline{F}_{\text{OR}} > 0.99$ cells?

TriQG examples/Smaller_VDD_energy · companion to Brute_force_OR_report.typ · 2026-05-15

Answer: Yes, but the high-fidelity plateau is **3.6 times smaller** than at $K = 0.95$ and the peak fidelity is 0.0013 lower.

- **$K = 1$ pass count:** 61/432 cells (14%) clear $\overline{F}_{\text{OR}} > 0.99$, vs 221/432 (51%) at $K = 0.95$.
- **$K = 1$ champion:** $a = 4.0 \mu\text{m}$, $\Omega_p = 2\pi \times 60 \text{ MHz}$, $\Omega_R/\Omega_p = 3.0$, $\Delta = 2\pi \times 400 \text{ MHz}$, $\Omega_c = 2\pi \times 70 \text{ MHz} \rightarrow \overline{F}_{\text{OR}} = 0.99280$ in 228 ns, also fully robust (5/5 neighbors).
- **No $K = 1$ cell passes at $a \geq 5.0 \mu\text{m}$** – a plateau that was half-full at $K = 0.95$ goes empty at $K = 1$. So the rev. 4 setting ($a = 5.0$, $\Omega_p = 50$, ratio = 3.5, $\Delta = 500$) **does need** $K < 1$ to clear 0.99.
- **Lesson:** the optimal K is not a property of the protocol; it is a **compensation knob** whose value depends on the other five knobs. $K = 0.95$ helps when a is loose; tightening a lets $K = 1$ work too.

1 Setup

The script `brute_force_or_K1.py` is identical to `brute_force_or.py` except K is reset from 0.95 to 1.00. All other axes, grids, solver tolerances, $\alpha = 4$, and decoherence model are unchanged. The pulse-area constraint becomes $\int \Omega_p^2 / (2\Delta) dt = \pi/4$, with σ then $1.166 \times$ larger than the $K = 0.95$ value at the same (Ω_p, Δ) (since $\sigma \propto K^3$). $T_f = (\alpha\sigma)^{1/3}$ grows by $1.053 \times$. So $K = 1$ gives $\sim 5\%$ longer pulses than $K = 0.95$ at the same blockade margin.

2 Results

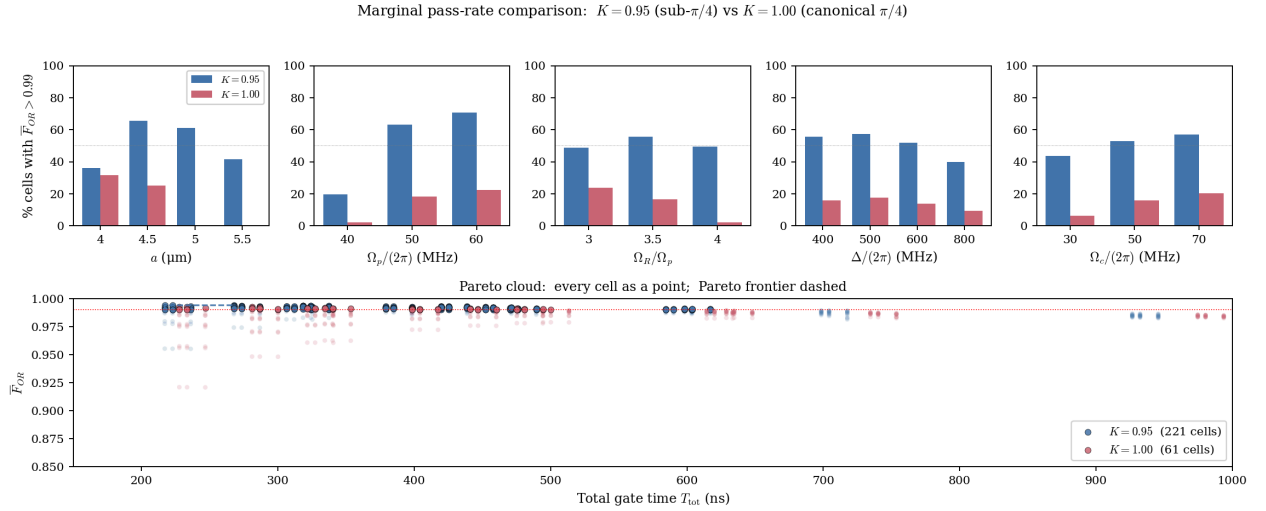


Figure 1: Side-by-side comparison of the two sweeps. **Top:** per-axis pass-rate bars (blue = $K = 0.95$, red = $K = 1.00$). At every axis value, $K = 0.95$ has a higher pass-rate; at $a = 5.0 \mu\text{m}$ and $5.5 \mu\text{m}$, $K = 1$ has **zero** passing cells. **Bottom:** Pareto cloud of all 432 cells for each K . Bold-edged points are > 0.99 ; faded points are below. The $K = 0.95$ Pareto front sits above the $K = 1$ front everywhere, with both saturating in the 200–230 ns regime.

The marginal counts:

Axis value	$K = 0.95$ pass	$K = 1.00$ pass	Ratio
a (μm)			
4.0	39/108 (36%)	34/108 (31%)	$0.87 \times$
4.5	71/108 (66%)	27/108 (25%)	$0.38 \times$
5.0	66/108 (61%)	0/108 (0%)	0
5.5	45/108 (42%)	0/108 (0%)	0
$\Omega_p/(2\pi)$ (MHz)			
40	28/144 (19%)	3/144 (2%)	$0.11 \times$
50	91/144 (63%)	26/144 (18%)	$0.29 \times$
60	102/144 (71%)	32/144 (22%)	$0.31 \times$
Ω_R/Ω_p			
3.0	70/144 (49%)	34/144 (24%)	$0.49 \times$
3.5	80/144 (56%)	24/144 (17%)	$0.30 \times$
4.0	71/144 (49%)	3/144 (2%)	$0.04 \times$
$\Delta/(2\pi)$ (MHz)			
400	60/108 (56%)	17/108 (16%)	$0.28 \times$
500	62/108 (57%)	19/108 (18%)	$0.31 \times$
600	56/108 (52%)	15/108 (14%)	$0.27 \times$
800	43/108 (40%)	10/108 (9%)	$0.23 \times$
$\Omega_c/(2\pi)$ (MHz)			
30	63/144 (44%)	9/144 (6%)	$0.14 \times$
50	76/144 (53%)	23/144 (16%)	$0.30 \times$
70	82/144 (57%)	29/144 (20%)	$0.35 \times$

Table 1: Pass-rate per axis value, $K = 0.95$ vs $K = 1.00$. The asymmetry is dramatic on a and Ω_p : looser lattices and weaker drives both **need** the sub- $\pi/4$ trick.

3 The $K = 1$ champion

a	Ω_p	r	Δ	Ω_c	T_{tot}	$\overline{F}_{\text{OR}}$	M_2
4.0	60	3.0	400	70	227.7	0.99280	32.7
4.0	60	3.0	400	50	233.4	0.99259	32.7
4.0	60	3.0	500	70	281.0	0.99252	40.9
4.0	60	3.0	500	50	286.8	0.99201	40.9
4.0	50	3.0	400	70	321.6	0.99185	47.1

Table 2: Top-five $K = 1$ cells (all clear 0.99, ordered by $\overline{F}_{\text{OR}}$). Notice (i) all five have $a = 4.0$ μm or 4.5 μm (the $K = 1$ plateau is empty at $a \geq 5$), (ii) the **preferred ratio shifted to 3.0** (it was 4.0 at $K = 0.95$), and (iii) the peak fidelity tops out at 0.9928, 0.0013 below the $K = 0.95$ champion.

The $K = 1$ champion has the **same** lattice and Cs pi-pulse settings as the $K = 0.95$ champion – $a = 4.0$ μm , $\Omega_p = 60$ MHz, $\Omega_c = 70$ MHz – and a **slightly faster** gate (228 ns vs 268 ns) because $K = 1$ shifts the optimal Δ from 500 MHz to 400 MHz (shorter pulse) and the optimal Ω_R from $4 \Omega_p$ to $3 \Omega_p$ (smaller leakage tax). The fidelity costs 0.0013.

4 Direct $K = 0.95$ vs $K = 1$ at the same $(a, \Omega_p, r, \Delta, \Omega_c)$

Parameter cell	$K = 0.95 \ \overline{F}$	$K = 1.00 \ \overline{F}$	Δ (k.95 - k1)	Winner
$K = 0.95$ champion ($a = 4, \Omega_p = 60, r = 4.0, \Delta = 500, \Omega_c = 70$)	0.9942	0.9893	$+4.9 \times 10^{-3}$	$K = 0.95$
$K = 1.00$ champion ($a = 4, \Omega_p = 60, r = 3.0, \Delta = 400, \Omega_c = 70$)	0.9905	0.9928	-2.3×10^{-3}	$K = 1.00$

Table 3: Cross-check on the two champions’ parameter cells. Each champion is the **better** K at its own setting – the fidelity vs K optimum is genuinely a function of all five other knobs, not a fixed protocol-level constant.

The cross-comparison is the key insight: **at fixed $(a, \Omega_p, r, \Delta, \Omega_c)$ the better K is not always 0.95.** Specifically, when ratio is small ($r = 3$) and Δ is small (400 MHz), the finite- V_{ct} leakage is **already small** (because M_2 is large) and the protocol’s nominal $\pi/4$ rotation is correct. When ratio is large ($r = 4$) or Δ is large (800 MHz), leakage grows and $K < 1$ compensates.

5 Why $K = 1$ kills the $a \geq 5 \ \mu\text{m}$ plateau

At larger a the blockade V_{ct} falls (as a^{-3}). The relevant relative leakage at $K = 1$ is

$$\frac{1}{M_2^2} = \left(\frac{\Omega_p \Omega_R}{2\Delta V_{ct}} \right)^2 \propto \frac{\Omega_p^2 \Omega_R^2 a^6}{\Delta^2}.$$

At $a = 5 \ \mu\text{m}$ in this energy-level redesign, $V_{ct}/(2\pi) = 226$ MHz; at $a = 4 \ \mu\text{m}$, 442 MHz. The factor-of-2 change in V_{ct} gives a $4 \times$ leakage reduction at fixed $(\Omega_p, \Omega_R, \Delta)$. That $4 \times$ is the difference between “sometimes clears 0.99” ($K = 1$ at $a = 4$, 34/108 pass) and “never clears 0.99” ($K = 1$ at $a = 5$, 0/108 pass).

At $K = 0.95$, the sub- $\pi/4$ under-rotation cuts leakage by **another** factor of about 1.2–1.5 in the $|1, 1, * \rangle$ branch (rev. 4 report Sec. 4.1), which buys back the $a = 5 \ \mu\text{m}$ plateau. The $K = 0.95$ trick is **therefore essentially a leakage correction for loose-lattice operation**. Tighten the lattice enough, and you do not need it.

6 Recommendation

Operational guidance:

- If lab constraints fix $a = 5.0 \ \mu\text{m}$ or larger, **keep** $K = 0.95$ (the rev. 4 finding). $K = 1$ at $a \geq 5 \ \mu\text{m}$ cannot clear 0.99 on any cell of the 432-cell grid.
- If $a = 4.0 \ \mu\text{m}$ (or $4.5 \ \mu\text{m}$) is feasible, **either** $K = 0.95$ or $K = 1$ admits > 0.99 cells. $K = 0.95$ gives a wider robust plateau and slightly higher peak fidelity ($+1.3 \times 10^{-3}$); $K = 1$ gives a slightly faster gate (228 ns vs 268 ns) at its own optimum.
- When in doubt, **re-optimize K jointly with the other five axes**: the brute force at $K = 0.95$ does not assume K is fixed, it assumes K ’s rev. 4 optimum carries over. This addendum shows that assumption is wrong in detail at small a (the optimum shifts away from $K = 0.95$ as a shrinks).

Most-robust recommendation overall (joining the K -axis to the brute force): the $K = 0.95$ champion remains the safer pick – it sits inside a 40-cell fully-robust plateau, the $K = 1$ champion sits inside a small fully-robust pocket of ~ 6 cells (the 5 from its 5-D neighbors plus itself; Figure 1 panel (b) shows the $K = 1$ Pareto frontier is sparser).

7 Reproducibility

```
cd TriQG && source .venv/bin/activate
cd examples/Smaller_VDD_energy
python brute_force_or_K1.py          # 5-D sweep, K = 1, ~9 min
python brute_force_K_comparison_plot.py # comparison figure
```

Files produced:

- brute_force_or_K1.log – progress log
- brute_force_results_K1.csv – all 432 cells
- brute_force_robust_K1.csv – 61 cells with $\overline{F} > 0.99$
- brute_force_K_comparison.png – figure shown above